

VERIFICATION REPORT

“DAK SRONG 2 HYDROPOWER PROJECT”

FINAL

**RINA**

Document Prepared By RINA Services S.p.A. (RINA)

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Summary:

RINA Services S.p.A. (RINA), commissioned by Kyoto Energy Pte. Ltd. has verified the greenhouse gas emission reductions reported for the project activity “Dak Srong 2 Hydropower Project” in Viet Nam, CDM Registration Reference N° 3389, for the period 30/10/2010 to 22/02/2011. The project was validated by Korea Energy Management Corporation (Validation Report No. GHGCC(A) 09-005 for the project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 04.4 of 22/11/2010) and it was registered as CDM Project activity on 23/02/2011. The proposed project activity aims to construct and operate a run-off river hydropower plant utilizing the water resources of the Ba river, which links to Da Nang river and eventually to the East Sea of Viet Nam. The electricity generated by the project is delivered to the Viet Nam national grid via a new single circle 110 kV transmission line. The project has a small run off reservoir, a weir, a penstock and a power house containing 3 turbines of 8290.2kW each and 3 generators of 8,000 kW each; which will generate electricity to replace the equivalent electricity supplied by the Viet Nam National grid dominated by the fossil fuel-fired power plants. The project activity will contribute to the GHG emission reductions and is expected to generate totally 44,466tCO₂e annually as estimated in the registered CDM PDD.

The purpose and scope of this verification is to ensure that reported emission reductions are complete and accurate in accordance with applicable UNFCCC criteria for CDM and VCS in order to be certified. The GHG emission reductions were calculated on the basis of the approved methodology ACM0002, version 11, “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” of 12/02/2010, and the monitoring plan included in the registered Project Description Document, version 3.6 of 19/06/2012.

During this verification, 1 Corrective Action Request (CAR) 4 Clarification Requests (CLs) were identified related to the monitoring, implementation or operations of the proposed VCS project activity (Monitoring Report version 01.0 of 04/04/2012) in relation to all relevant VCS and UNFCCC requirements and all relevant host Party criteria and the applied baseline and monitoring methodology, and these CARs and CLs are successfully closed after necessary corrections/clarifications by the client.

In conclusion, it is RINA's opinion that the project activity “Dak Srong 2 Hydropower Project”, in Viet Nam, as described in the latest Monitoring Report version 01.1 of 04/12/2012, meets all relevant requirements for VCS/CDM activities and all relevant host Party criteria and correctly applies the baseline and monitoring methodology ACM0002, version 11, “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” of 12/02/2010. Hence RINA is able to certify that the emission reductions from the project during the monitoring period 30/10/2010 to 22/02/2011 amount to 16,200 tCO₂e.

Table of Contents

1	Introduction	5
1.1	Objective	5
1.2	Scope and Criteria	5
1.3	Level of assurance	6
1.4	Summary Description of the Project.....	6
2	Validation Process, Findings and Conclusion.....	7
2.1	Validation Process	7
2.2	Validation Findings.....	9
2.2.1	Gap Validation.....	9
2.2.2	Methodology Deviations	11
2.2.3	Project Description Deviations	11
2.2.4	New Project Activity Instances	11
2.3	Validation Conclusion.....	11
3	Verification Process.....	11
3.1	Method and Criteria.....	11
3.2	Document Review	12
3.3	Interviews.....	14
3.4	Site Inspections.....	14
3.5	Resolution of Any Material Discrepancy.....	14
4	Verification Findings	17
4.1	Project Implementation Status.....	17
4.2	Accuracy of GHG Emission Reduction or Removal Calculations	17
4.3	Quality of Evidence to Determine GHG Emission Reductions or Removals.....	25
4.4	Management and Operational System.....	26

5	Verification conclusion	28
	APPENDIX A.....	30

1 INTRODUCTION

1.1 Objective

RINA has been commissioned by “Kyoto Energy Pte. Ltd.” to perform an independent verification of its VCS project, “Dak Srong 2 Hydropower Project”, already registered under CDM with CDM Project ID. 3389 /c/ for the reported GHG emission reductions for the given monitoring period 30/10/2010 up to 22/02/2011 being the period between the start of the project activity and the registration of the project under CDM. The VCS projects must undergo independent third party verification and certification of emission reductions as the basis for issuance of Voluntary Emission Reductions (VERs/VCUs).

The objectives of this verification exercise are, by review of objective evidence, to establish that:

- The project activity has been implemented and operated as per the project description (PD) and that all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place;
- Monitoring report and other supporting documents are complete;
- The data is recorded and stored as per the monitoring methodology and approved monitoring plan.
- To confirm that the monitoring system is implemented and fully functional to generate Voluntary Emission Reductions (VERs/VCUs) without any double counting, and

To establish that the data reported are accurate, complete, consistent, transparent and free of material error or omission by checking the monitoring records and the emissions reduction calculation.

1.2 Scope and Criteria

The verification scope is:

- to verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;
- to evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emission reduction data is free from material misstatement;
- to verify that reported GHG emission data is sufficiently supported by evidence.

The project is assessed against the requirements of VCS version 3 and related rules and guidance. RINA has, based on the recommendations in the latest version of CDM Validation and Verification Manual, and employed a rule-based approach (as criteria) in the verification, focusing on the identification of significant reporting rules and the reliability of project monitoring.

Verification is not meant to provide any consultancy towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the monitoring.

1.3 Level of assurance

All the revisions of the verification report before being submitted to the client were subjected to an independent internal technical review to confirm that all verification activities had been completed according to the pertinent RINA instructions. The technical review was performed by a technical reviewer(s) qualified in accordance with RINA's qualification scheme for CDM/VCS validation and verification.

The verification team and the technical reviewers consist of the following personnel:

Role	Last Name	First Name	Country
Team Leader / Verifier	Arokiasamy	Cyril Augustus	India
Technical Expert	Mathew	Vijay	India
Technical reviewer	Menon	Rekha	India

1.4 Summary Description of the Project

As per the registered PDD version 3.5 of 10/11/2010 and the revised PDD version 3.6 of 19/06/2012 addressing the post registration changes^{*/cl*}, the proposed project activity aims to construct and operate a run-off river hydropower plant utilizing the water resources of the Ba river, which links to Da Nang river and eventually to the East Sea of Viet Nam. The generated electricity is exported to the Viet Nam national grid. The project has a small run off reservoir, a weir, a penstock and a power house containing 3 turbines of 8290.2kW each and 3 generators of 8,000 kW each. The generated electricity is exported to the Vietnam national grid through a new 110 kV single circuit transmission line from 110 kV Transformer Station of the Dak Srong 2 to 110 kV transformer station of the Dak Srong 2 A hydropower plant and then connect through the 110 kV transformer station of Dak Srong hydropower plant to the National Grid. RINA conducted the site visit from 17/04/2012 to 19/04/2012 and confirmed that all the above mentioned equipments are in place and is in operation. The capacities of the equipments were also verified by the verification team during the site visit by checking the equipment name plates. The same was also cross verified by cross checking the certificate provided by Central Power Corporation; the Viet Nameese national grid, stating that the plant started its operation since 30/10/2010^{*/il*}. Verification team also confirms that the project activity is implemented at one site. All the three turbine/gensets were commissioned at the same time and hence no phased implementation has happened. It was also noted that the plant has operated normally throughout the monitored period. No data or variables in the monitoring report are different from the registered PDD, which could cause an increase in estimates of the emission reductions in the current monitoring period or is highly likely to increase the estimates of emission reductions in the future monitoring periods.

Project Participant(s)	Hoang Anh Gia Lai Hydropower Joint Stock Company		
Project Title	Dak Srong 2 Hydropower Project		
Location of the project	Latitude of 13°41'45"N and longitude of 108°33'30"E		
Methodology(ies)	"ACM0002", "Consolidated Methodology for Grid Connected Electricity Generation from Renewable Sources", version 11 of 12/02/2010/ b//		
Sectoral Scope(s)	01	RINA's Technical Area(s)	1.2
Registered PDD	Revision 3.5 of 10/11/2010		
Date of registration	23/02/2011	CDM Registration Reference N°	3389
Revised monitoring plan	Revision 3.6 of 19/06/2012		
Starting date of the crediting period	23/02/2011		
Project's crediting period	23/02/2011 to 22/02/2018		
Monitoring period	30/10/2010 to 22/02/2011		
Project documentation link	http://cdm.unfccc.int/Projects/DB/KEMCO1265792796.76/view		
Purpose of the project activity	The project activity consists of the installation of three turbine generators of 8000 kW each; resulting a total capacity 24 MW. The purpose of the project activity is to generate electricity from renewable hydro power and exports to the Viet Nameese national grid, thereby leading to the reduction of GHG emissions by displacing the equivalent amount of electricity which would have been generated predominantly from fossil fuel fired power projects connected to the grid as indicated in the registered PDD.		

2 VALIDATION PROCESS, FINDINGS AND CONCLUSION

2.1 Validation Process

Method and Criteria: Validation was conducted using RINA procedures in line with the requirements specified in the VCS Requirements, i.e. VCS Version 3 Requirements Document, dated 04/10/2012

Document Review: The VCS Project Description, version 01.0 of 04/04/2012, version 01.1 of 04/12/2012/a/, was assessed as part of the validation. The following table lists the documentation that was reviewed during the validation.

/a/	Kyoto Energy Pte. Ltd.: VCS-PD for project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 1 of 04/04/2012; version 01.1 of 04/12/2012
/b/	CDM Executive Board: Baseline and monitoring methodology “ACM0002”, “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”, version 11 of 12/02/2010
/c/	Kyoto Energy Pte. Ltd.: Registered CDM-PDD for project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 3.5 of 10/11/2010 , revised PDD version 3.6 of 19/06/2012 Linkage: UNFCCC website: http://cdm.unfccc.int/PRCCContainer/DB/prcp671086109/view English language and retrieved on 27/11/2012
/d/	VCSA, VCS Standard: VCS Program Guide, v3.4 dated 04/10/2012
/e/	VCSA, VCS Program Guide: VCS Standard, v3.3 dated 04/10/2012
/f/	VCSA: Registration and Issuance Process: VCS Procedural Document, version 3.4 of 04/10/2012
/g/	VCSA: Program Definitions: VCS Program Definitions, v3.4 dated 04/10/2012
/h/	VCSA: VCS Project Description Template, VCS Version 3, v3.1 dated 04/10/2012
/i/	Central Power Corporation: Letter dated 30/11/2010 confirming the commercial operation date of Dak Srong 2 Hydropower Project
/j/	Korea Energy Management Corporation: Validation Report No. GHGCC(A) 09-005 for the project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 04.4 of 22/11/2010 Linkage: UNFCCC website http://cdm.unfccc.int/PRCCContainer/DB/prcp671086109/view , English language and retrieved on 27/11/2012
/k/	HAGL Hydropower JSC: PPA signed with Central Power Company dated 27/05/2010
/l/	Hoang Anh Gia Lai Hydropower Joint Stock Company: LoA from Ministry Of Natural Resources And Environment Of Viet Nam (DNA) dated 30/12/2009.
/m/	VCS: VCS Guidance, Validation and Verification Manual, v3.0, dated 04/10/2012

Interviews: The key personnel interviewed and the main topics of the interviews are summarized in the table below.

	Date	Name and Role	Organization	Topic
1	17/04/2012	Mr. Nguyen Le Anh Duy – Director	Dak Srong 2 – HAGL Hydro power Joint Stock Company	Project implementation, commercial operation date of the plant, payment details from EVN
2		Mr. Dinh Duy Khoi –		Monitoring procedures, data

		Site Supervisor		recording, training provided, emergency preparedness
3		Mr. Pham Manh Hung - Operator		Data management, day to day record handling.
4	17/04/2012 and 19/04/2012	Mrs. Nguyen Thi Lan – project Manager		Data monitoring, transfer of data, archiving of data, operation and maintenance of the plant, internal and external audit
5	17/04/2012, 18/04/2012 and 19/04/2012	Mr. Arijit Paul – Regional Manager	Kyoto Energy Pte Ltd.	Data transfer, ER calculations, monitoring report.
6		Mr. Nguyen Manh Khuong – CDM Project Executive		Data transfer, ER calculations, monitoring report

Site Inspections: From 17/04/2012 to 19/04/2012, RINA visited the Dak Strong 2 Hydropower Project site, which is situated at Yang Nam, Ya Ma, and Dak Hninh Communes Kong Chro district, Gia Lai Province, Socialist Republic of Viet Nam for the purpose of CDM verification of the project activity. During the on-site assessment of the project RINA assessed the implementation and operation of the proposed project activity, accessed the installed equipment directly, interviewed key personnel of the plant to confirm the operation information, the Project Start Date, the GHG Programs undertaken. RINA did not carry out a separate site visit for this VCS project as monitoring period for CDM was post the monitoring period of the proposed VCS verification.

Resolution of Any Material Discrepancy: The objective of this phase of the validation is to resolve any outstanding issues which need to be clarified for RINA's positive conclusion on the project description. To guarantee transparency any findings raised regarding to the validation are incorporated in the Verification Protocol and Verification Protocol Tables in Appendix A to this report (Referred to section 3.5 of this report).

2.2 Validation Findings

2.2.1 Gap Validation

In according with the requirements of section 3.11.8 of VCS Standard, v3.3 /e/, RINA performed a gap validation for the project's compliance with the VCS rules, as the proposed project activity has been registered (UNFCCC Ref. No. 3389)/**cl**/ under an approved GHG program – CDM program under Kyoto Protocol.

Sectoral Scope and Project Type:

According to the VCS Version 3 Requirements Document and the list of Sectoral Scopes of the UNFCCC, and confirmed by review of the registered CDM PDD/**cl**/ and the applied CDM methodology ACM0002/**bl**/, the project is not a grouped project and is applicable under the following sectoral scope:

- Sectoral Scope 01: Energy industries (renewable-/non-renewable sources)

Project Start Date and Project Crediting Period:

The start date of the project activity was on 30/10/2010 when the project plant started commissioning and connected to the grid for electricity generation*/il* and the CDM emission reduction start date was on the date of CDM registration, i.e. 23/02/2011*/cl*. The project was commissioned on 30/10/2010 and that date represents the date from which the emissions reductions under VCS will be claimed. Therefore RINA confirmed that the project crediting period for VCS verification is from 30/10/2010 to 22/02/2011.

The start date of this monitoring period is 30/10/2010 which is the project start date and the end date of this monitoring period is 22/02/2011 which is prior to the CDM crediting period. RINA thus confirmed that the monitoring period for VCS verification is within the valid crediting period under VCS regulation.

Based on the information available on UNFCCC website*/cl*, the project has been registered with the renewable crediting period from 23/02/2011 to 22/02/2018 and the validation report was completed and firstly issued by Korea Energy Management Corporation on 22/11/2010*/jl*.

Project Scale and Estimated GHG Emission Reductions or Removals:

The proposed project activity was categorized as **Project**, since the estimated average annual GHG emission reductions is 44,466tCO₂e, less than 1,000,000tCO₂e per year as per the registered CDM PDD*/cl*.

Project Location:

The project is located in Yang Nam, Ya Ma, and Dak Hninh Communes Kong Chro district, Gia Lai Province, Socialist Republic of Viet Nam. The geographic coordinates of the project is 108° 33'30" east longitude and 13°41'45" north latitude. This has been crosschecked by GPS measures during site visit.

The project boundary has been identified in the registered CDM PDD*/cl* as the Vietnam national grid system and the power plants connected to it; which mainly comprises a range of thermal, gas, diesel oil and hydropower plants. The physical site visit confirmed that the project boundary in place is consistent with the registered CDM PDD*/cl* and the registered validation report*/jl*.

Conditions Prior to Project Initiation:

Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to Calculate the Emission Factor for an Electricity System". These conditions have been validated by validation DOE in the registered CDM validation report */jl*.

Ownership and Other Programs:

Proof of Title: The owner of the emission reductions generated by the proposed project is Hoang Anh Giai Lai Hydropower Joint Stock Company. The proof of title is verified from the commissioning certificate // and Letter of Approval (LoA) for CDM issued by the Viet Nam DNA//.

Emissions Trading Programs and Other Binding Limits: The project activity contributes to reductions in the emissions of CO₂ by renewable energy generation, thus displacing electricity generation based on fossil fuels in the EVN (Electricity Viet Nam) grid. It was confirmed by the PP that no rejection was received by other GHG programs. The CDM Program is a GHG program approved by the VCS program. Emission reductions till the CDM registration date will be claimed by PP under the VCS programme. Therefore RINA confirmed that the GHG emission reductions generated by the project before CDM registration date have not and will not be used in the emissions trading program or for the purposed of demonstrating compliance with the binding limits that are in place in that jurisdiction or sector.

The proposed project activity has been registered under CDM programs of Kyoto Protocol. The registration number is 3389/c/. There is no other form of GHG-related environmental credit generated by the proposed project activity. The same has been confirmed from the UNFCCC website and CDM validation report/tj/.

Commercially Sensitive Information:

All information in the project documents is available for the public review, as claimed by PP in the VCS Project Description.

2.2.2 Methodology Deviations

There is no methodology deviations applied to the project.

2.2.3 Project Description Deviations

There is no project description deviations applied to the project.

2.2.4 New Project Activity Instances

The project is not a grouped project. Hence this section is Not Applicable.

2.3 Validation Conclusion

RINA is of opinion that the proposed project activity confirms to the paragraph 3.2 of validation and verification manual version 3.0/m/..

3 VERIFICATION PROCESS

3.1 Method and Criteria

Verification was conducted using RINA procedures in line with the requirements specified in the VCS Requirements, CDM M&P, the latest version of the CDM Validation and Verification Manual, and relevant decisions of the COP/MOP and the CDM EB and applying standard auditing techniques. The GHG emission reductions are on the basis of the approved Baseline and monitoring methodology “ACM0002”, “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”, version 11 of 12/02/2010.

The verification consisted of the following three phases:

- Document review;
- On-site assessment including Interviews and Site Inspections;
- Resolution of Any Material Discrepancy and the issuance of the final verification report and certification.

The following sections outline each step in more detail.

3.2 Document Review

The monitoring report (MR), version 01 of 04/04/2012, version 01.1 of 04/12/2012/**03/**, the emission reduction calculations spreadsheet (VCS VER-Dak Srong 2 VCS MR Ver 1.0) version 01 submitted on 06/11/2012/**04/**, were assessed as part of the verification. In addition the registered Project Design Document (CDM PDD)/**01/** in particular the baseline estimations and the monitoring plan, and the registered CDM validation report, revision 4.4 of 22/11/2010/**09/** for the project were reviewed.

The following table lists the documentation that was reviewed during the verification.

/01/	Kyoto Energy Pte. Ltd.: Registered CDM-PDD for project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 3.5 of 10/11/2010 , revised PDD version 3.6 of 19/06/2012 Linkage: UNFCCC website: http://cdm.unfccc.int/PRCCContainer/DB/prcp671086109/view English language and retrieved on 27/11/2012
/02/	Kyoto Energy Pte. Ltd.: VCS-PD for project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 1 of 04/04/2012; version 01.1 of 04/12/2012
/03/	Kyoto Energy Pte. Ltd.: VCS-MR for project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 1 of 04/04/2012; version 1.1 04/12/2012
/04/	Kyoto Energy Pte. Ltd.: Emission reduction calculation spreadsheet (VCS VER-Dak Srong 2 VCS MR Ver 1.0) version 01 submitted on 06/11/2012.
/05/	VCSA, VCS Standard: VCS Program Guide, v3.4 dated 04/10/2012 VCSA, VCS Program Guide: VCS Standard, v3.3 dated 04/10/2012
/06/	VCSA, VCS Standard: Validation and Verification manual, v3.0 dated 04/10/2012
/07/	CDM Executive Board: Baseline and monitoring methodology “ACM0002”, “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”, version 11 of 12/02/2010
/08/	Area of the reservoir measured in the surface of the water calculation sheet along with the project design submitted on 27/12/2012.
/09/	Korea Energy Management Corporation: Validation Report No. GHGCC(A) 09-005 for the project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 04.4 of 22/11/2010

/10/	Central Power Corporation: Letter dated 30/11/2010 confirming the commercial operation date of Dak Srong 2 Hydropower Project on 30/10/2010
/11/	Central Electrical Testing Company: Calibration certificates dated 06/10/2010 1. For the main meters with serial number 10020491, 10020492 and 10020493 vide certificate numbers 06105692/BB-ETC-P6, 06105694/BB-ETC-P6 and 06105696/BB- ETC-P6 respectively 2. For the backup meters with serial number 10018220, 10018221 and 10018222 vide certificate numbers 06105693/BB-ETC-P6, 06105695/BB-ETC-P6 and 06105697/BB-ETC-P6 respectively
/12/	Song Da electrical engineering Joint Stock Company: Calibration test certificate for the TEG meters with serial numbers 09090155620006, 09090155620007 and 09090155620008 dated 10/10/2010 vide certificate serial numbers TND/DSR2-01, TND/DSR2-02 and TND/DSR2-03
/13/	HAGL Hydropower JSC: PPA signed with Central Power Company dated 27/05/2010
/14/	Central Electrical Testing Company: Calibration certificates dated 25/07/2011 1. For the main meters with serial number 10020491, 10020492 and 10020493 vide certificate numbers 06115119/CPC ETC-DL, 06115121/CPC ETC-DL and 06115123/CPC ETC-DL 2. For the back up meters with serial number 10018220, 10018221 and 10018222 vide certificate numbers 06115120/CPC ETC-DL, 06115122/CPC ETC-DL and 06115124/CPC ETC-DL
/15/	VCS: Monitoring report template (v3.2), VCS version 03.0 dated 04/10/2012
/16/	Kyoto Energy Pte Ltd.: CDM Monitoring report for project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 1.0 of 14/03/2012, version 02 of 21/05/2012, version 03 of 19/06/2012 and version 03.2 of 27/06/2012 related to the monitoring period 23/02/2011 to 29/02/2012.
/17/	RINA: Verification report for the project activity “Dak Srong 2 Hydropower Project” in Viet Nam, version 1.2 of 14/08/2012, for the monitoring period 23/02/2011 to 29/02/2012 http://cdm.unfccc.int/Projects/DB/KEMCO1265792796.76/view
/18/	Ministry of Natural Resources and Environment and Ministry Of Finance, Government of Viet Nam: Circular No: 58/2008/TTLT-BTC-BTN&MT
/19/	IPCC, 2006 IPCC Guidelines for National Greenhouse Gas Inventories
/20/	Study, Definition of Vietnam grid emission Factor, Department of Meteorology, Hydrology and Climate Change, 12/2009
/21/	Department of Meteorology, Hydrology and Climate Change: Final report on Study, Definition of Viet Nam Grid Emission Factor” published in December 2009
/22/	Elster: Technical specification for the meters with model number A1700 CT metering
/23/	Ministry of Industry: Regulation on “Technical requirements of measurement equipments in the power plant” issued on 09/01/2007
/24/	Ministry of Science and Technology: Decision No. 25/2007/Q Đ -BKHCN published on 05/10/2007 regarding the calibration procedure and frequency
/25/	EVN receipts of electricity exported and imported for the project Dak Srong 2 for the months of October – 2010, November – 2010, December-2010, January -2011 and February-2011.
/26/	HAGL Hydropower JSC: Monthly log book for TEG meter for the months October 2010 to February 2012
/27/	Wasion Group: Technical specification for the meters with model number DSSD331

3.3 Interviews

Interviews: The key personnel interviewed and the main topics of the interviews are summarized in the table below.

	Date	Name and Role	Organization	Topic
1	17/04/2012	Mr. Nguyen Le Anh Duy – Director	Dak Srong 2 – HAGL Hydro power Joint Stock Company	Project implementation, commercial operation date of the plant, payment details from EVN
2		Mr. Dinh Duy Khoi – Site Supervisor		Monitoring procedures, data recording, training provided, emergency preparedness
3		Mr. Pham Manh Hung - Operator		Data management, day to day record handling.
4	17/04/2012 and 19/04/2012	Mrs. Nguyen Thi Lan – project Manager		Data monitoring, transfer of data, archiving of data, operation and maintenance of the plant, internal and external audit
5	17/04/2012, 18/04/2012 and 19/04/2012	Mr. Arijit Paul – Regional Manager	Kyoto Energy Pte Ltd.	Data transfer, ER calculations, monitoring report.
6		Mr. Nguyen Manh Khuong – CDM Project Executive		Data transfer, ER calculations, monitoring report

3.4 Site Inspections

From 17/04/2012 to 19/04/2012, RINA visited the Dak Srong 2 Hydropower Project site, which is situated at Yang Nam, Ya Ma, and Dak Hninh Communes Kong Chro district, Gia Lai Province, Socialist Republic of Viet Nam for the purpose of CDM verification of the project activity. CDM verification was carried out by RINA and the team leader for the VCS validation and verification (the project activity) remains the same as that for the CDM verification. During the on-site assessment of the project RINA assessed the implementation and operation of the proposed project activity, accessed the installed equipment directly, interviewed key personnel of the plant to confirm the operation information, the Project Start Date, the GHG Programs undertaken. RINA did not carry out a separate site visit for this VCS project as monitoring period for CDM was post the monitoring period of the proposed VCS verification.

3.5 Resolution of Any Material Discrepancy

The objective of this phase of the verification is to resolve any outstanding issues which need to be clarified for RINA's positive conclusion on the monitoring report and emission reductions.

To guarantee transparency a verification protocol has been customized for the project. The protocol shows in a transparent manner the requirements, means of verification and the results from verifying the identified criteria.

The verification protocol consists of three tables; the different columns in these tables are described in the figure below (see Figure 1). The completed verification protocol is enclosed in Appendix A to this report.

A corrective action request (CAR) is raised if one of the following occurs:

- Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impair the estimate of emission reductions;
- Issues identified in a FAR during validation to be verified during verification have not been resolved by the project participants.

A clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.

A forward action request (FAR) is raised during verification for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CARs, CLs and FARs identified are included in the verification protocol in Appendix A of this report.

Figure 1 Verification protocol tables

Verification Protocol, Table 1 - Requirement checklist				
Checklist Question	Ref.	MoV	Comments	Conclusion
The various requirements in Table 1 are linked to checklist questions the project should meet. The checklist is organized in seven different sections.	Makes reference to documents where the answer to the checklist question or item is found.	Explain how conformance with the checklist question is investigated. Examples are document review (DR), interview or any other follow-up actions (I), cross checking (CC) with available information relating to projects, (N/A) means not applicable.	The discussion on how the conclusion is arrived at and the conclusion on the compliance with checklist question so far.	OK is used if the information and evidence provided is adequate to demonstrate compliance with VCS requirements. For CAR, CL and FAR see the definitions above.

Verification Protocol, Table 2 - Resolution of Corrective Action Requests and Clarification			
Corrective action requests and/or clarification requests	Reference to Table 1	Response by project participants	Verification conclusion
The CAR and/or CLs raised in table 1 are repeated here.	Reference to the checklist question number in Table 1 where the CAR or CL is explained.	The responses given by the project participants to address the CARs and/or CLs.	The verification team's assessment and final conclusion of the CARs and/or CLs.

Verification Protocol, Table 3 - Forward Action Requests		
Forward action request	Reference to Table 1	Response by project participants Verification conclusion
The FAR raised in table 1 is repeated here.	Reference to the checklist question number in Table 1 where the FAR is explained.	Response by the project participants on how forward action request will be addressed.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

RINA confirmed that the project implementation is in line with the description in registered CDM PDD/**01/** by site visit conducted for the purpose of CDM verification of the project activity. RINA performed a physical site visit for the CDM verification of the project activity from 17/04/2012 to 19/04/2012 to verify the real implementation of the project against the description in its registered CDM PDD/**01/** and found that 3 generating units each of capacity 8 MW were in place of the project site and on operation. The project has been located in Yang Nam, Ya Ma, and Dak Hninh Communes, Kong Chro District, Gia Lai Province of Viet Nam. RINA confirmed that three generating units were commissioned and the project is supplying electricity to Viet Nam National Grid and the implementation is in line with the description in registered PDD/**01/**. The project started operation on 30/10/2010 as per the letter from Central Power Corporation regarding commercial operation date of Dak Srong 2 Hydropower Project/**10/** and then connected to the Viet Nam National grid as per the power purchase agreement signed/**13/**. The description is also consistent with the real implementation confirmed as above. Therefore RINA confirmed that there is no material discrepancy between the project implementation and the project description in the PDD.

According to the registered CDM monitoring plan/**01/**, six energy meters (3 mains and 3 backups) were installed with the project activity, which belonging to the grid operator EVN. The main meters are of accuracy class 0.2s and backup meters are of accuracy class 0.5s. There are 3 meters of accuracy class 0.5s located after the Generator 1, Generator 2 and Generator 3 respectively. They measure the total electricity generation of each generator. RINA confirms that the monitoring plan would result in a conservative monitoring and calculation for the emission reductions in section 4.2 below of this report.

During the CDM site visit, verification team noted that one bidirectional energy meter was installed for each turbo generator to monitor the electricity exported to the grid and the electricity imported from grid. A back up meter is also installed to ensure correct functioning of the energy meter. This was not consistent with the registered CDM PDD which stated that one electricity meter would be installed for total generation of the facility and one for electricity consumed from the grid at this facility **/17/**.

Verification team further noted that the actual monitoring plan and installed equipment goes beyond the requirements of the PDD and does not result in decrease in accuracy of monitoring. However PP has notified CDM EB on the change in monitoring plan. The registered PDD has also been revised (version 3.6) accordingly and issuance has been done based on the revised PDD **/01/**.

Based on the review of registered CDM validation report **/09/**, no FAR was raised during the validation and/or there was no FAR to be addressed during previous verification since this is the only one Pre-CDM VCS verification before the CDM crediting period.

4.2 Accuracy of GHG Emission Reduction or Removal Calculations

According to the applied methodology “ACM0002”, “Consolidated baseline methodology for grid connected electricity generation from renewable sources”, version 11 of 12/02/2010/06/, the emission reductions have been calculated based on the following formula:

$$ER_y = BE_y - PE_y - L_y$$

Where,

ER_y : Emission reductions in year y (tCO₂e/yr)

BE_y : Baseline emissions in year y (tCO₂e/yr)

PE_y : Project emissions in year y (tCO₂e/yr)

L_y : Leakage in year y (tCO₂e/yr)

The baseline emissions (BE_y) for the project activity are calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

$$EG_{PJ,y} = EG_{facility,y}$$

Where,

$EG_{facility,y}$: Quantity of net electricity generation supplied by the project plant/unit to the grid in year y

$EF_{grid,CM,y}$: Combined margin emission factor for grid connected power generation in year calculated ex-ante, as 0.5764 tCO₂ /MWh/01/

Combined margin emission factor was fixed ex-ante at the time of validation. The value was cross checked with the registered PDD and found to be consistent.

As per the methodology, for hydro power project activities that result in new reservoirs, project proponents shall account for CH₄ and CO₂ emissions from the reservoir if the power density of the project activity (PD) is greater than 4 W/m² and less than or equal to 10 W/m².

Accordingly PP has calculated the power density based on the following equation as provided in the applied methodology:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where,

PD : Power density of the project activity

Cap_{PJ} : Installed capacity of the hydropower plant after the implementation of the project activity

Cap_{BL} : Installed capacity of the hydropower plant before the implementation of the project activity (W). For new hydropower plants, this value is zero

A_{PJ} : Area of the reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoirs are full

A_{BL} : Area of the reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoirs are full (m²). For the new reservoir, this value is zero

Power density is thus calculated to be 5.67W/m² which is in between 4W/m² and 10W/m². Thus PP has calculated the project emissions as follows:

$$PE_{HP,y} = \frac{EF_{Res} * TEG_y}{1000}$$

Where,

PE_{HP,y} : Emission from reservoir expressed as tCO₂e/year

EF_{Res} : The default emission factor for emissions from reservoirs in kg CO₂e/MWh

TEG_y : Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads in year y expressed in MWh/yr

As per the methodology and the registered PDD, the project does not need consider leakage.

Hence E_{Ry} = B_{Ey} - P_{Ey}

i.e. E_{Ry} = 19,278 – 3,078 = **16,200** tCO₂e

The following parameters were available at validation as per the registered CDM PDD/**01**/ and the validation report/**09**/:

DATA/PARAMETER	Source of data	Reported value for the project period	Assessment/Observation
Combined margin CO ₂ emission factor for grid connected power generation as calculated in DNA issued "Study, Definition of Viet Nam Grid Emission Factor, 2009" (EF	As per the "Tool to calculate the emission factor for an electricity system".	0.5764 tCO ₂ /MWh	The verification team has verified the value with the report published by DNA in December 2009/ 21 / and registered PDD/ 01 / and found to be correct and consistent. Hence accepted.

grid,CM,y)			
Default emission factor for emission from reservoirs (EF Res)	Decision by EB 23	90kg CO ₂ /MWh	The verification team has verified the value with the registered PDD/01/ and the methodology applied/07/ and found to be correct and consistent. Hence accepted.
Global warming potential of methane valid for the relevant commitment period (GWPC _{H4})	IPCC	21 tCO ₂ e/tCH ₄ for the first commitment period	The verification team has verified the value with the registered PDD/01/ and found to be correct and consistent. Hence accepted.

Data and parameters monitored ex-post:

DATA/PARAMETER	EG _{facility,y}
Data Unit	MWh
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid during the current monitoring period
Source of data to be used	MM1, MM2, MM3; BM1,BM2, BM3 in case of the MM failure
Value data for the monitoring period	33,445. This is the sum of net export as measured from meters installed for all the three units (unit 1, unit 2 and unit 3).
Measuring frequency	Continuous measuring. The same is in line with the methodology requirement and is hence accepted.
Reporting frequency and recording procedure	Monthly recording. As per the monitoring report, version 01.1 of 04/12/2012/03/ and the monitoring plan of the final CDM PDD/01/, the energy meters of each unit measures the export and import of the electricity generated. The representatives from EVN (Viet Nam national grid; Central Power Corporation), PP and Gia Lai province electricity company take down the monthly reading on first day of every month. This is then recorded in the EVN receipt and is signed by these representatives. Invoices are raised by PP based on these receipts and accordingly payment is done by EVN. EVN further cross checks the main meter reading with back up meter reading so as to ensure that the reading recorded from the main meter is correct. Verification team has performed a site visit for CDM verification of the project activity; during the site visit the same has been checked and confirms that the same procedure is followed at site.
Type of monitoring equipment	The monitoring equipments used for the three turbo generators are bi directional digital meters as seen at site and the details of the meters are as follows.

	Meter	Tag	Meter No.	Model	Manufacturer
	Main meter	MM 1	10020491	A1700	Elster
	Main meter	MM 2	10020492	A1700	Elster
	Main meter	MM 3	10020493	A1700	Elster
	Back Up Meter	BM 1	10018220	A1700	Elster
	Back Up Meter	BM 2	10018221	A1700	Elster
	Back Up Meter	BM 3	10018222	A1700	Elster
Is accuracy of the monitoring equipment as stated in the PDD?	0.2s and 0.5s for Main and back up meters respectively. The same was confirmed by the verification team during the site visit and by cross checking and verifying the calibration certificates /11/14/22/. The accuracy class was selected by the PP based on the regulation No.02/2007/QD-BCN on "Technical requirements of measurement equipments in the power plant" issued by Ministry of Industry on 09/01/2007/23/.				
Calibration frequency/interval	Meters will be calibrated yearly.				
Is the calibration interval in line with the monitoring plan of the PDD?	As per the monitoring plan, version 01.1, the meters will be calibrated annually. No frequency has been mentioned in the monitoring plan of the registered PDD.				
How were the values in the monitoring report verified and cross-checked?	As per the registered CDM PDD, the electricity export should have been cross checked with the sales receipt/invoices. PP has submitted the sales receipt/invoices for the monitoring period/25/. The validation team has verified the document and found that the some value mentioned in the document is not matching with the values used for the calculation. However the PP has justified that discrepancies in the value used (less than 0.50 % during the whole monitoring period) are due to rounding off error. The validation team has accepted the justification, as the paragraph 5.3.1 of the VCS Standard (v3.4) states that, the materiality thresholds for VCS projects is 5% (paragraph 2.9.1 of the VCS Standard). The validation team confirms that the change in value is not more than 5% and the value used is not on the higher side. Since the approach chosen is conservative, verification team accepted the same.				
Does the data management (from monitoring equipment to emission	The initial calibration test for both main and back up meters were conducted on 06/10/2010. As the test				

reduction calculation) ensure correct transfer of data and reporting of emission reductions?	certificate provided a validity of 1 year; the monitoring period fall under this period. Verification team had checked the certificates and found appropriate/11/. It was also noted that the meters are well within their accuracy limit.
If only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not Applicable.

DATA/PARAMETER	TEG _y																								
Data Unit	MWh																								
Description	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads during the monitoring period																								
Source of data to be used	TEG 1, TEG 2 and TEG 3 meters																								
Value data for the monitoring period	34,195																								
Measuring frequency	Continuous measuring. The same is in line with the methodology requirement and is hence accepted.																								
Reporting frequency and recording procedure	Monthly recording. The same is in line with the methodology requirement and is hence accepted.																								
Type of monitoring equipment	The monitoring equipments used as seen at site (which is conducted for CDM verification) are as follows. <table><tr><th>Meter</th><th>Tag</th><th>Meter No.</th><th>Model</th><th>Manufacturer</th></tr><tr><td>TEG meter</td><td>TEG 1</td><td>0909015 5620006</td><td>DSSD331</td><td>Wasion</td></tr><tr><td>TEG meter</td><td>TEG 2</td><td>0909015 5620007</td><td>DSSD331</td><td>Wasion</td></tr><tr><td>TEG meter</td><td>TEG 3</td><td>0909015 5620008</td><td>DSSD331</td><td>Wasion</td></tr></table> As per the monitoring report, version 01.1 of 04/12/2012 /03/ and the monitoring plan of the final CDM PDD/01/, the total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, during the monitoring period is monitored by installing an electricity meter at project activity site. A monthly log book is maintained at site in order to record the measurements displayed at these meters/26/.					Meter	Tag	Meter No.	Model	Manufacturer	TEG meter	TEG 1	0909015 5620006	DSSD331	Wasion	TEG meter	TEG 2	0909015 5620007	DSSD331	Wasion	TEG meter	TEG 3	0909015 5620008	DSSD331	Wasion
Meter	Tag	Meter No.	Model	Manufacturer																					
TEG meter	TEG 1	0909015 5620006	DSSD331	Wasion																					
TEG meter	TEG 2	0909015 5620007	DSSD331	Wasion																					
TEG meter	TEG 3	0909015 5620008	DSSD331	Wasion																					

	Verification team has confirmed the same during the site visit conducted for CDM verification for the project activity.
Is accuracy of the monitoring equipment as stated in the PDD?	0.5s. The same was confirmed by the verification team during the site visit conducted for the CDM verification of the project activity; and by cross checking and verifying the calibration certificates/12/. The same accuracy class was also mentioned in the technical specification provided by manufacturer/12/27/.
Calibration frequency/interval	Meters will be calibrated once in 2 years.
Is the calibration interval in line with the monitoring plan of the PDD?	As per the monitoring plan the meters will be calibrated once in 2 years, which is in line with the decision No. 25/2007/QD-BKHCN dated 05/10/2007 /23/. No frequency has been mentioned in the monitoring plan of the registered CDM PDD.
How were the values in the monitoring report verified and cross-checked?	The PP has submitted the monthly log book details to cross check the values used for the parameter/26/. The validation team found the document submitted is appropriate.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions?	The initial calibration test was conducted on 10/10/2010 and as the frequency is fixed for once in 2 years as the government rule, the monitoring period fall under this period. Verification team had checked the certificates and found appropriate/12/. It was also noted that the meters are well within their accuracy limit.
If only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not Applicable

DATA/PARAMETER	Cap _{PJ}
Data Unit	W
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data to be used	Project site (Nameplates on generators at project site)
Value data for the monitoring period	24,000,000
Measuring frequency	Yearly. The same is in line with the methodology requirement and is hence accepted.
Reporting frequency and recording procedure	Yearly. The same is in line with the methodology requirement and is hence accepted.
Type of monitoring equipment	Not applicable

Is accuracy of the monitoring equipment as stated in the PDD?	Not applicable
Calibration frequency/interval	Not applicable
Is the calibration interval in line with the monitoring plan of the PDD?	Not applicable
How were the values in the monitoring report verified and cross-checked?	The values have been verified during the site visit conducted for the CDM verification of the project activity.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions?	Not applicable
If only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not applicable

DATA/PARAMETER	A_{PJ}
Data Unit	m^2
Description	Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m^2).
Source of data to be used	Project site
Value data for the monitoring period	4,236,000
Measuring frequency	Yearly. The same is in line with the methodology requirement and is hence accepted.
Reporting frequency and recording procedure	Yearly. The same is in line with the methodology requirement and is hence accepted.
Type of monitoring equipment	Water level gauge on the side of the reservoir.
Is accuracy of the monitoring equipment as stated in the PDD?	Not applicable, as by water level gauge it means a scale marked in the wall of the reservoir.
Calibration frequency/interval	Not applicable
Is the calibration interval in line with the monitoring plan of the PDD?	Not applicable
How were the values in the monitoring report verified and cross-checked?	The values have been verified during the site visit conducted for the CDM verification of the project activity. The same has been crosschecked with the calculation sheet provided by the PP/08/.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct	Not applicable

transfer of data and reporting of emission reductions?	
If only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not applicable

Emission Reductions Achieved:

The emission reductions calculation reported in the VCS Monitoring Report **/03/** and ER calculation spreadsheet (ERCS)**/04/** have been verified to be correct and in line with the CDM PDD**/01/**. The emission reductions by the project activity for the monitoring period from 30/10/2010 to 22/02/2011 are equivalent to 16,200 tCO₂e, as reported in the VCS monitoring report**/01/**. The data presented in the VCS MR/03/ and ERCS**/04/** was assessed by reviewing in detail project documentation, collection of monitored data, observation of established monitoring and reporting practices and assessment of the reliability of monitoring equipment. Sufficient evidence was presented and verified by RINA for the reported emission reductions as listed in the above.

RINA also compared the actual emission reductions with the estimates in the registered CDM PDD **/01/** as follows:

The reported emission reductions are equivalent to 16,200 tCO₂e (on average 141 tCO₂e per day) during the monitoring period from 30/10/2010 to 22/02/2011 (totally 115 days). According to the registered CDM PDD**/01/**, the estimated emission reductions are equivalent to 44,466 tCO₂e annually, on average 122 tCO₂e per day. The reported averaged emission reductions are 15% higher than the estimated average emission reductions. However according to the historical monthly discharge on 38 years of The Ba River at the Dak Srong 2 dam site, the total percentage of annual discharge at the site from beginning of November to the end of January is 44.08%, corresponding to 19,600 tCO₂e based on the yearly estimated PDD. Comparing with this value with the actual emission reduction value for the proposed monitoring period (16,200 tCO₂e), the actual emission reduction is lesser than the estimated. The validation team has also compared the emission reduction in the month of October -2011, November-2011, December-2011, January-2012 and February-2012 from the first CDM verification report for the project activity for the monitoring period 23/02/2011 to 29/02/2012**/17/** and found that the generation for the said period is comparable with the value of the proposed VCS monitoring period.

Hence, RINA confirmed that the actual emission reductions reported during the monitoring period are less than the corresponding estimates in the registered CDM PDD**/01/**.

4.3 Quality of Evidence to Determine GHG Emission Reductions or Removals

Quantity of net electricity generation supplied by the project plant/unit to the grid is measured with the help of the bidirectional meter installed at site for each turbo generators. Representatives

from EVN (Viet Nam Electricity; Central Power Corporation), PP and Gia Lai province Electricity Company take down the monthly reading on first day of every month. This is then recorded in the EVN receipt and is signed by these representatives. Invoices are raised by PP based on these receipts and accordingly payment is done by EVN. Emission reduction is calculated based on the readings taken from the main meter and recorded in the EVN receipt. Verification team has checked the same and found to be in order. Hence it is confirmed that the calculation method is proper as the readings are sourced from a document approved by EVN who is the buyer of the electricity generated. EVN further cross checked the main meter reading with back up meter reading and ensured that the reading recorded from the main meter is correct and accordingly payment was released as per the invoice raised by PP. Verification team has checked the ER calculation and confirmed that the values are correctly taken from the EVN receipts/25/.

CO₂ emission factor has been taken as 0.5764 tCO₂ /MWh. The verification team has verified the value with the report published by DNA in December 2009/20/ and registered CDM PDD/01/ and found to be correct and consistent. The baseline emission is calculated as 19,278.

As the power density is 5.67W/m² which is in between 4W/m² and 10W/m², project emission is involved in the project activity and is calculated to be 3,078 tCO₂ e/year.

As per the methodology and the registered PDD, the project does not need consider leakage.

Thus the emission reduction is calculated to be 16,200 tCO₂ e for this monitoring year/04/.

4.4 Management and Operational System

Verification team has done the site visit from 17/04/2012 to 19/04/2012 for CDM verification of the project activity and cross-checked the procedures being followed at site in monitoring and calculating the electricity generation exported to the grid. The team confirmed that PP follows the procedures as mentioned in the registered PDD/01/. 3 main meters are installed at site so as to monitor the net electricity exported to grid by the project activity. This is then cross checked with the back-up meter readings installed at site so as to ensure the correctness of the data. Further the main meters and backup meters will be will be calibrated annually; and the TEG meters will be calibrated once in 2 years as per decision No. 25/2007/QD-BKHCN dated 05/10/2007/24/ or at the end of the validity period of the calibration result (if specifically mentioned in the calibration certificate) whichever is earlier. As the calibration test certificate submitted by PP for the main and back up meters specifies a validity of one year, which ensures the proper working of the meters /11/. Verification team has checked the monthly log book maintained at site for the energy meters and calibration certificates; and confirmed that proper measures were taken by PP to ensure the appropriateness of the value considered in the ER calculation /11/12/26/. Hence verification team concluded that PP has a proper management system so as to ensure the correct recording and reporting of data used for the emission reduction calculations.

A monitoring team has been established for this project activity, which is headed by a CDM project manager, who will be conducting external audit for the project activity. The verification

team has confirmed the same during the site visit. The documents related to the monitoring will be archived up to two years after the end of the crediting period or last issuance of this project activity whichever comes later.

As per the VCS MR, version 01.1 dated 04/12/2012/**03/**, a formal set of monitoring procedures has been followed which is as per the registered CDM PDD/**01/** and VCS PD/**02/** that includes the details of the organization, control and steps required for certain key monitoring system features including VCS staff training, VCS data and record keeping arrangements, data collection, VCS data quality control and quality assurance, equipment maintenance, equipment calibration and equipment failure. The following are the details of the monitoring responsibilities allocation.

	Tasks description (and frequency)	Operator	Supervisor	Site manager / Project director	CDM consultant Project manager
Monitoring activity					
1	Recording of manual data	✓			
Quality Assurance & Quality Control					
2	Verification of data monitored (consistency and completeness)		✓		
3	Ensuring adequate training of staff		✓		
4	Ensuring adequate maintenance		✓		
	Ensuring calibration of monitoring instruments		✓		
5	Data archiving: ensuring adequate storage of data monitored (integrity and backup): 2 years after the end of the crediting period			✓	
6	Identification of non-conformance and corrective/preventive actions and monitoring plan improvement		✓		
7	Emergency procedures		✓		
8	External audit				✓
Calculation of GHG emission reductions and reporting					
9	Processing of data and calculation of emission reductions			✓	
10	Monitoring report: management review of monitoring report (internal audit)			✓	

During the CDM verification site visit; the verification team found that these were followed on the site. Training records for the monitoring personnel were reviewed during CDM verification site visit. The training would be provided to the concern personnel as required. Thus, they are competent to carry out the relevant tasks with sufficient accuracy. All necessary monitored and measured raw data were checked during on-site verification.

5 VERIFICATION CONCLUSION

RINA Service S.p.A (RINA) has performed verification of the emission reductions reported for the project activity “Dak Srong 2 Hydropower Project” in Viet Nam, CDM Registration Reference N° 3389, for the period 30/10/2010 to 22/02/2011, with regard to the relevant requirements for VCS and CDM rules.

The project participants of the “Dak Srong 2 Hydropower Project” project are responsible for:

- The preparation of greenhouses gas emissions data and the reported greenhouse gas emission reductions from the project on the basis set out in the monitoring plan contained in the registered CDM project design document version 3.6 of 19/06/2012
- The development and maintenance of records and reporting procedures in accordance with that plan, including the calculation and determination of greenhouse gas emission reductions of the project

It is the responsibility of RINA to express an independent verification opinion about the project's conformity with the VCS requirements and procedures and on the reported greenhouse gas emission reductions from the project.

Based on documented evidence and corroborated by an on-site assessment RINA can confirm that:

- The project has been implemented and operated as per the registered CDM-PDD;
- The monitoring plan in the registered CDM-PDD is as per the applied baseline and monitoring methodology.
- The monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS and CDM requirements;
- The emission reductions for the monitoring period are conservative based on the monitoring deviation;

It is RINA's opinion that the GHG emission reduction stated in the VCS monitoring report version 01.1 of 04/12/2012 for the “Dak Srong 2 Hydropower Project” in Viet Nam for the period 30/10/2010 to 22/02/2011 are fairly stated. The GHG emission reductions were calculated correctly on the basis of the baseline and monitoring methodology ACM0002, version 11,

“Consolidated baseline methodology for grid-connected electricity generation from renewable sources” of 12/02/2010.

Hence RINA is able to certify that the emission reductions from the project during the monitoring period 30/10/2010 to 22/02/2011 amount to 16,200 tCO₂e.

Reporting period: From [30/10/2010] to [22/02/2011]

Verified GHG emission reductions or removals in the above reporting period:

GHG Emission Reductions or Removals	tCO ₂ e
Baseline Emissions	19,278
Project Emissions	3,078
Leakage	0
Net GHG emission reductions or removals	16,200

APPENDIX A

VERIFICATION PROTOCOL

TABLE 1 REQUIREMENTS CHECK LIST

Checklist Question		Reference	MoV ¹	Comments	Conclusion
A Monitoring Report					
A.1	Does the used project title clearly enable the reader to identify the unique VCS activity? Is there an indication of a revision number, the date of the revision and the monitoring period?	/01//02//03/	DR/CC	Yes; the title of the project activity is unique. The Monitoring report also provides the revision number i.e. version 1.0 and date of the report is 4/4/2012. The monitoring period for the project activity is 30/10/2010 to 22/02/2011.	OK
A.2	Does the project comply with the applicable requirements for completing the Monitoring Reports (latest version available)?	/01//03//05/ /15/	DR	Yes; the project complies with the applicable requirements for completing the Monitoring Report. The Monitoring report is in line with VCS Program Guide Version 03 dated 04/10/2012, VCS Standard Version 03 dated 04/10/2012 and VCS: Monitoring report template (v3.2), VCS version 03.0 dated 04/10/2012.	OK
A.3	Does the MR comply with the template available (latest version)?	/01//03//15/	DR	The PP has used the latest MR template available.	OK
B Description of Project Activity					
A.1	Title of the project activity, revision number and date of Monitoring Report	/01//02//03/	DR	Title of the project activity is "Dak Srong 2 Hydropower Project" version number 1.0 dated 04/04/2012 and found to be consistent with the VCS PD and registered CDM PDD.	OK
A.2	Is the actual implementation and operation of the proposed project activity in accordance with the project activity in the registered VCS-PD?	/01//02//03/ /17/	DR/I	The project activity is a run-off river hydro power plant which utilizes the hydrological resources of Ba river which is located in the central highland province which joins with the Da Rang river to run into	OK

¹ MoV: DR document review, I interview, CC cross checking

Checklist Question		Reference	MoV ¹	Comments	Conclusion
				the East Sea. The project involves the installation of three turbo generator of 8 MW each totalling to 24 MW which was confirmed by the verification team by checking the name plate fixed in the equipments at site. The electricity thus generated will be exported to Viet Nam national grid. Verification of the CDM project for the monitoring period 23/02/2011 to 29/02/2012 was conducted by RINA and a site visit was carried out for this purpose. Hence a separate site visit for the Validation and verification under VCS has not been carried out as the monitoring period under this project activity is prior to that of the monitoring period under CDM for which the issuance is already done. The onsite visit for the CDM verification was conducted on 17/04/2012, 18/04/2012 and 19/04/2012 The site were accessible to all the locations of the hydro power plant without facing any barriers. Thus confirmed that the turbo generators have been commissioned and started operation since 30/10/2010. The same was cross checked with the letter provided by Central Power Corporation (CPC- a part of EVN)	
A.3	Methodology applied for the registered project activity	/01//02//03/ /07/	DR/CC	Project activity applies the methodology ACM0002, "Consolidated baseline methodology for grid-connected electricity generation from renewable sources", version 11 of 12/02/2010. This methodology is applicable to the project activity as it involves the installation of a new hydro power unit with a small run off river reservoir and the power density of the power plant is greater than 4W/m ² .	OK
B Monitoring					
B.1 Monitoring plan					
B.1	Is the actual implementation and operation of the	/01//02//03/	DR/CC	The project activity is registered under the Clean	OK

Checklist Question		Reference	MoV ¹	Comments	Conclusion
	proposed project activity in accordance with the project activity in the registered VCS-PD?	/17/	/I	Development mechanism of the UNFCCC; and the actual implementation and operation of the project activity is in accordance with the registered PDD and CERs has been issued for the monitoring period 23/02/2011 to 29/02/2012.	
B.2	In case of deviation between the registered project and the actual implementation/operation, do they comply with the requirements of the Project Standards?	/01//02//03//17/	DR/CC /I	Please refer section B.1	OK
B.3	For project activity that consist of more than one site: describe the status of the implementation and starting date of operation of each site; For project activity with phased implementation: describe the progress of the proposed project activity achieved in each phase number; if the phased implementation is delayed, described the reasons and the expected implementation dates.	/01//02//03//04/	DR/CC /I	The project activity is at Gia Lai Province, Yang Nam, Ya Ma, and Dak Hninh Communes Kong Chro District, Viet Nam. The physical location has been confirmed during the site visit. The project activity is in operation stage.	OK
B.4	Methodology and methodological tool applied for the registered project activity	/01//02//03//04//07/	DR/CC	The methodology used for the project activity is ACM0002 "Consolidated baseline methodology for Grid Connected Electricity Generation from Renewable Sources", version 11, (which is available at UNFCCC website). The same is consistent with PDD registered under the Clean Development mechanism of the UNFCCC.	OK
C Compliance of the monitoring activities with the registered monitoring plan / Compliance of the monitoring plan with the monitoring methodology and methodological tool					
C.1 Monitoring plan					
C.1.1	Does the monitoring plan included in the registered VCS project activity comply with the applied methodology?	/01//02//03//04//07/	DR/CC	As per the applied methodology, monitoring shall consist of metering the net electricity generation supplied to the grid by the project plant, total electricity produced by the project activity, including the electricity supplied to grid and the electricity supplied to the internal loads, installed capacity of the hydro power plant after the	OK

Checklist Question		Reference	MoV ¹	Comments	Conclusion
				implementation of the project activity and the area of the reservoir measured in the surface of water, after the implementation of project activity, when the reservoir is full. The monitoring plan of the registered PDD is transparent in monitoring the above mentioned parameters; and hence the monitoring plan is in accordance with the applied methodology.	
C.1.2	Does the monitoring comply with the monitoring plan in the registered VCS-PD?	/01//02//03/	DR/CC	Yes; the monitoring complies with the monitoring plan in the registered CDM PDD.	OK
C.2 Data and parameters fixed ex-ante or at renewal crediting period					
C.2.1	Which parameters were available at validation and how were they verified?	/01//02//03/ /19//20/	DR/CC	The parameters that were available at the time of validation where $EF_{grid,CM,y}$, EF_{Res} and GWP_{CH4} . The CO_2 emission factor of grid was calculated as per the "tool to calculate emission factor for an electricity system" and the values are sourced from the report published by the DNA of Viet Nam "Study, Definition of Viet Nam Grid Emission Factor, 2009". Emission factor of reservoir is taken as the default value based on the decision by EB 23. GWP_{CH4} is the default value taken from IPCC. Verification team checked the values mentioned in the monitoring report with that indicated in the registered PDD and confirmed to be appropriate.	OK
C.2.2	What default data were selected and applied?	/01//02//03/	DR/CC	$EF_{grid,CM,y}$: Combined margin emission factor for grid connected power generation in year calculated ex-ante, as 0.5764 tCO₂ /MWh EF_{Res}: Default emission factor for emission from reservoirs (EF_{Res}), considered as 90 kgCO₂/MWh GWP_{CH4} :Global warming potential of methane used for the project activity is not mentioned as a parameter	CAR-1 OK

Checklist Question	Reference	MoV ¹	Comments	Conclusion
			available at validation in the Monitoring report as available in the registered PDD	
C.3 Data and parameters monitored ex-post				
C.3.1 Which parameter have been monitored during the monitoring period? (Data/Parameter monitored / Data unit / Description / Source of data to be used / Value data for the monitoring period)	/01//02//03/	DR/CC	Data/ Parameter monitored: EG facility,y: Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh. The value is sourced from the EVN receipt prepared based on the monthly reading taken from the main meters installed at site. The value for the current monitoring period is 33,445 MWh. PP is requested to submit the copies of invoice raised by the PP to EVN and the evidence of payment received from EVN for the monitoring period. Data/ Parameter monitored: TEG_y: Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y measured in MWh. This value is taken from the energy meters installed just after the each generator. The value for the current monitoring period is 34,195 MWh. PP is requested to submit the copies of invoice raised by the PP to EVN and the evidence of payment received from EVN for the monitoring period. <i>PP is requested to submit copies of the log book maintained at plant for the relevant period to cross check the above details.</i> Data/ Parameter monitored: A_{PJ}: Area of the reservoir measured in the surface of the	CL-1 OK

Checklist Question	Reference	MoV ¹	Comments	Conclusion
			water, after the implementation of the project activity, when the reservoir is full in m². The value for this monitoring period is 4,236,000m². The value was calculated based on the water level measurements taken by PP with the help of a water level gauge installed in the reservoir. PP is requested to submit the evidence for the water level measurements for the monitoring period. Data/ Parameter monitored: Cap_{PJ}: Installed capacity of the hydro power plant after the implementation of the project activity. Value stated in monitoring report is 24,000,000 W. The equipment specification is available on the nameplate of the equipment issued by the supplier. Consistent with site visit observation, Hence accepted.	
C.3.2 Is the measurement equipment described? Is the accuracy of the measurement equipment addressed and deemed appropriate?	/01//02//10/ /11//14/	DR/CC	The details such as the serial number, manufacturer, model, accuracy class, date of calibration and its frequency of the main meters, back up meters and the meters installed after the generators are provided in the monitoring report. All the main and back up meters are of model A1700 manufactured by Elster. The accuracy class of main meters are 0.2s and that of back up meter and the TEG meter which are the meters installed after the turbine generators are of 0.5s. TEG meters are manufactured by Wasion and are of model DSSD331. Date of calibration for main and back up meters are 06/10/2010 and for TEG meter, it is 10/10/2010. Calibration frequency has been selected as yearly for main meters and back up meters; and once in 2 years for TEG meters. The serial numbers of all	CL-2 OK

Checklist Question		Reference	MoV ¹	Comments	Conclusion
				meters are as follows: Main meter 1 (MM1) : 10020491 Main meter 2 (MM2) : 10020492 Main meter 3 (MM3) : 10020493 Back up meter 1 (BM1) : 10018220 Back up meter 2 (BM2) : 10018221 Back up meter 3 (BM3) : 10018222 TEG meter (TEG1) : 09090155620006 TEG meter (TEG2) : 09090155620007 TEG meter (TEG3) : 09090155620008 The calibration covers the monitoring period. The date of calibration of the main meters and check meters are on 06/10/2012. The date of calibration of TEG meters are 10/10/2012 <i>However MR is not clear on the following:</i> <i>The reason for selecting the calibration frequency as in yearly main meters and check meters; once in 2 years for TEG meters and how it is in accordance with PPA signed.</i> <i>Entity responsible for conducting the calibration.</i>	
C.3.3	Is the measuring/reading/recording frequency adequate for all monitoring parameters? Is it in line with the registered monitoring plan?	/02/	DR/CC /I	The monitoring frequency for EG_{facility,y} and TEG_y is continuous. The monitoring for the parameter A_{PJ} and Cap_{PJ} are conducted yearly. The same was confirmed by the verification team during the site visit and is in line with the registered monitoring plan.	OK
C.4 Calibration requirements					
C.4.1	Are the requirements for maintenance and calibration of measurement equipment described and deemed	/01//02//03/	DR/CC	Please see the section C.3.2	CL-2

Checklist Question		Reference	MoV ¹	Comments	Conclusion
	appropriate?		/I		OK
C.4.2	Does the calibration cover the monitoring period?	/01//03//10/ 11//14/	DR/CC	Please see the section C.3.2	CL-2 OK
C.4.3	Has the calibration frequency been respected?	/01//02//03/ /10//11//14/	DR/CC	Please see the section C.3.2	CL-2 OK
C.4.4	In case of delay, describe the applied maximum permissible error	/01//02//03/	DR/CC	There is no delay. Hence no need of applying the maximum permissible error.	OK
C.5 Monitoring of the sustainable indicators					
C.5.1	Is the monitoring of sustainable development indicators/ environmental impacts warranted by legislation in the host Country?	/01//02//03/	DR	Yes; Circular No: 58/2008/TTLT-BTC-BTN&MT issued by the Ministry of Natural Resources and Environment and by the Ministry Of Finance, Government of Viet Nam, 1.2% of the CER revenue from the implementation of Renewable Energy Sources projects will be given to the Vietnam Environment Protection Fund (VEPF). However PP is requested give evidence on the same.	CL-3 OK
C.6 Management system and quality control					
C.6.1	How has it been assessed that the monitoring arrangements described in the monitoring plan are feasible within the project design?	/01//02//03/	DR/ I/CC	Site visit has done for CDM verification on 17/04/2012, 18/04/2012 and 19/04/2012; and checked the procedures followed at site in monitoring and calculating the net electricity exported to the grid. During the site visit validation team has confirmed that the monitoring arrangements are feasible within the project design.	OK
C.6.2	Are procedures identified for day-to-day record handling (including what records to keep, storage area of records and how to process performance documentation)?	/01//02//03/	DR/I/C C	MR clearly mention the procedures for daily internal record handling; and PP will keep all monitoring data required for verification and issuance for two years after	OK

Checklist Question		Reference	MoV ¹	Comments	Conclusion
	Will all monitored data required for verification and issuance be kept for two years after the end of the crediting period or the last issuance of CERs, for this project activity, whichever occurs later?			the end of the crediting period, for this project activity.	
C.6.3	Are the data management and quality assurance and quality control procedures sufficient to ensure that the emission reductions achieved by/resulting from the project can be reported ex post and verified?	/01//02//03/	DR// /CC	Yes. The data management, quality assurance and quality control procedures followed at site ensure that the emission reductions achieved by/resulting from the project can be reported ex post and verified. PP had conducted a review internally and made sure that there were no major data inconsistencies for this particular monitoring period. Please see section C.3.1	CL-1 OK
C.6.4	Are the responsibilities and authorities for monitoring and reporting in accordance with the responsibilities and authorities stated in the monitoring plan?	/01//02//03/	DR// /CC	Yes; the responsibilities and authorities for monitoring and reporting are in accordance with the responsibilities and authorities stated in the monitoring plan.	OK
C.6.5	Does data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions?	/01//02//03/	DR// /CC	The data management procedure is transparent; which will ensure correct transfer of data and reporting of emission reductions.	OK
D.1 Assessment of data and calculation of emission reductions/Accuracy of emission reduction calculations					
D.1.1	How were the values in the monitoring report verified and cross-checked?	/01//02//03/	DR// CC	Please see section C.3.1	CL-1 OK
D.1.2	If only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	/01//02/	DR// CC	Please see section C.3.1	CL-1 OK
D.1.3	Emission reductions reported	/01//02//04/	DR// CC	The emission reduction reported for the monitoring period is 16,200 tCO ₂ e. Please see section C.3.1	CL-1

Checklist Question	Reference	MoV ¹	Comments	Conclusion
				OK
D.1.4 Difference between the emission reductions estimated in the registered PDD and the emission reductions reported for the monitoring period.	/01//02//04/	DR// CC	As per the registered PDD the annual emission reduction is estimated as 44,466tCO ₂ e. The monitoring period is from 30/10/2010 to 22/02/2011. The emission reduction estimated during the monitoring period is 16,200 tCO ₂ e. PP is requested to justify the increase in emission reduction. Also please see section C.3.1	CL1 CL4 OK

TABLE 2 RESOLUTION OF CORRECTIVE ACTION REQUESTS AND CLARIFICATION REQUESTS

Corrective action and/ or clarification requests	Reference to Table 2	Response by project participants	Verification Conclusion
CAR 1 GWP_{CH4} :Global warming potential of methane used for the project activity is not mentioned as a parameter available at validation as available in the registered PDD	C.2.2	The MR is now updated accordingly.	PP has revised the MR and the revision made is found appropriate. Hence accepted. CAR is closed.
CL 1 - PP is requested to submit the copies of invoice raised by the PP to EVN and the evidence of payment received from EVN for the monitoring period. - PP is requested to submit copies of the log book maintained at plant for the monitoring period.	C.3.1, C.6.3, D.1.1, D.1.2, D.1.3, D.1.4	<u>1st response by PP (28/11/2012)</u> - Please refer to «SD 01a_Invoice» as the copies of invoice raised by the PP to EVN, and «SD 01b-Payment» are the bank transfer from EVN to the PO for the monitoring period. - Please refer to «SD 02_Internal monthly record» as the copies log book maintained signed by the site manager for the monitoring period <u>2nd response by PP (18/12/2012)</u> The small discrepancies (less than 0.50 % during the whole monitoring period) are due to rounding off error. As per the paragraph 5.3.1 of the VCS Standard (v3.4), the materiality thresholds for VCS projects is 5% (paragraph 2.9.1 of the VCS Standard). Because the identified error is less than the VCS materiality threshold, no change on the ER sheet have been implemented.	PP has submitted the invoice and copies of log book maintained. However the values of the invoice is not matching with the values provided in the ER calculation spread sheet. CL is open. Justification provided by PP is acceptable. Hence CL 1 is closed.

Corrective action and/ or clarification requests	Reference to Table 2	Response by project participants	Verification Conclusion
		Please refer to «SD 09_EVN receipts», «SD 10_Feb 2011_Daily generation»	
CL 2 MR is not clear on the following: - The reason for selecting the calibration frequency as in yearly main meters and check meters; once in 2 years for TEG meters and how it is in accordance with PPA signed. - Entity responsible for conducting the calibration.	C.3.2, C.4.1, C.4.2, C.4.3	1. For main meters and backup meters the calibration frequency has been kept as yearly since the validity of the calibration certificates issued for these meters are one year. Please refer to the relevant calibration certificates as «SD 03_Calibration» for the main and back up meters for this monitoring period. Kindly note that the PPA does not mention any specific calibration frequency but it requires the PP to comply with the relevant regulation in relation to calibration frequency of energy meters. Please refer to «SD 04_PPA_DS2» (point 9) and «SD 05_Regulation_Energy_Meters». Please note the regulation requires calibration every 2 years. Hence by having annual calibration the PP goes beyond the requirement of the PPA provisions. For TEG meters, the calibration certificates does not specifically provide the validity of the certificates. Thus the calibration frequency has been fixed as per the regulation. (SD 05_Regulation_Energy_Meters). 2. The calibration of the 3 main meters and 3 backup meters were conducted by the Central Electrical Testing Company	The justification provided by PP is found appropriate. The validation team has reviewed the supportive documents and found acceptable. Hence CL 2 is closed.

Corrective action and/ or clarification requests	Reference to Table 2	Response by project participants	Verification Conclusion
		which is a subsidiary of EVN (Viet Nam Electricity). The calibration of the 3 TEG meters conducted by Song Da Electrical Engineering Joint Stock Company, which holds business lines is installing, testing, maintaining, and upgrading telecommunication system, electric stations, grids, transformer stations and technological lines. (refer to SD 08_SDEE JSC)	
CL 3 PP is requested to submit evidence for the fulfilment of the requirement against Circular No: 58/2008/TTLT-BTC-BTN&MT issued by the Ministry of Natural Resources and Environment and by the Ministry Of Finance, Government of Viet Nam for the monitoring of sustainable development indicators warranted by legislation in the host Country.	C.5.1	Kindly note tha the demonstration of compliance with circular Circular No: 58/2008/TTLT-BTC-BTN&MT can only be completed when CERs for Dak Srong 2 Hydropower project are issued. Meanwhile, please see «SD 06_LoA» for confirmationfrom the Ministry of Natural Resources and Environment (MONRE), Vietnam that the project positively contributes towards the sustainable development goals of Viet Nam.	The justification provided by PP is found appropriate. The validation team has reviewed the supportive documents and found acceptable. Hence CL 3 is closed.
CL 4 As per the registered PDD the annual emission reduction is estimated as 44,466tCO₂e. The monitoring period is from 30/10/2010 to 22/02/2011. The emission reduction estimated during the monitoring period is 16,200 tCO₂e. PP is requested to justify the increase in emission reduction.	D.1.4	The proposed monitoring period happened during the rainy season. Indeed, according to the historical monthly discharge on 38 years of The Ba River at the Dak Srong 2 damsite, the total percentage of annual discharge at the site from beginning of November to the end of January is 44.08%, corresponding to 19,600 tCO ₂ e based on the yearly estimated PDD.	The justification provided by PP is found appropriate. The validation team has reviewed the supportive documents and found acceptable. Hence CL 4 is closed.

Corrective action and/ or clarification requests	Reference to Table 2	Response by project participants	Verification Conclusion
		Please note that the months of October and of February have conservatively not been included for determining the percentage of the annual discharge. Comparing with the actual emission reduction value for our monitoring period (16,200 tCO₂e), the actual emission reduction is smaller than the estimated. Hence, there is no excess of emission reductions to report. Please refer to: «SD 07a_Historical monthly discharge» and «SD 07b_Equivalent ex-ante ERs»	

TABLE 3 FORWARD ACTION REQUEST

Forward action request	Reference to Table 2	Response by project participants Verification Conclusion
FAR 1		